

PREPARED FOR

T6N, R4W, Mount Diablo Meridian

COORDINATES 38.3459°N 122.3047°W WGS84	ELEVATION 66 ft NAVD88 · MSL	STATE PLANE CA 2 (0402) · N 1,887,922.81 E 6,474,289.05 (US ft, NAD83)	FEMA ZONE Zone X 06055C0510FF	DECLINATION 13.0° E WMM 2026
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APN 038050009000 | ZONING AP | ACRES 0.481 ac

BEST GNSS WINDOW Mon, Jul 13 · 6:00 AM – 10:00 AM | PDOP 0.99 · 34 sv | NEAREST CORS
P261 · 14.1 mi ✓

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Site file · Google Earth
Download KMZ

RECORDER
Napa County Recorder-Clerk · 1127 First
St, Napa, CA 94559

NEAREST CORS
P261 · 14.1 mi

GENERATED
July 13, 2026

ASSESSOR
Napa County Assessor · 1127 First St,
Napa, CA 94559

NEAREST BM
JT9010 · 0.2 mi

SYSTEM
TopoScout · info@toposcout.com

This report compiles data from public and third-party sources (NGS, FEMA, USGS, NOAA, and county records) current as of the generation date shown above. GNSS planning values are predictive and conditions may change. This document is provided for planning purposes only and does not constitute a boundary survey, legal land description, or engineering certification. Verify all critical values against primary sources.

SITE & SIGNAL REPORT

COORDINATES

38.3459, -122.3047

WGS84 / NAD83

ELEVATION

66 ft

Above MSL (approx)

MAGNETIC DECLINATION

13.0° E

Compass reads 13.0° too far W

FEMA FLOOD ZONE

X

AREA OF MINIMAL FLOOD HAZARD · Panel 06055C0510FF

IONOSPHERIC CONDITIONS

PEAK KP INDEX

4.7

AVG KP

2.4

SOLAR FLUX F10.7

138 sfu

TEC (VTEC) ▲ DEGRADED

20.9 TECU

-1.8 TECU vs median

ASSESSMENT

MODERATE ACTIVITY

Peak Kp 4.7 (avg 2.4). Solar flux F10.7 = 138 sfu. Moderate geomagnetic activity. Some ionospheric delay possible. Static sessions should use longer occupation times.

GNSS / DOP SUMMARY

DATE	AVG PDOP	BEST 4-HR WINDOW	AVG SATS	RATING
Mon, Jul 13	0.99	6:00 AM – 10:00 AM	34	Ideal

GPS + GLONASS + Galileo + BeiDou MEO · Full multi-constellation model · 15° elevation mask · Per-day independent computation
GNSS satellite geometry is accurate within a 50-mile radius of the entered location – orbital mechanics do not change meaningfully at ground-level distances. Weather data is specific to the entered coordinates.

WILDFIRE PROXIMITY

No Active Fires Within 150 Miles

No active fires as of report date (July 13, 2026) · Conditions may change – verify before fieldwork

GPS CONSTELLATION HEALTH

▲ 1 Satellite Currently Unhealthy – PRN 11

Active outage per NAVCEN · Affected satellites auto-excluded by receiver

 Land Cover & Multipath Risk

SKY OBSTRUCTION

MULTIPATH RISK

Minor Obstruction

Low vegetation or sparse development nearby — minor sky obstruction possible.

Low Multipath

Minor reflective surfaces present — multipath generally manageable.

RECOMMENDED ELEVATION MASK

13° Minor vegetation — slight horizon reduction

✓ PDOP Check at 13°: 0.8–0.8 across survey window
PDOP remains ≤ 0.8 at 13° — geometry confirmed acceptable

5-POINT SAMPLE (CENTER + N/S/E/W · ~100 M)

Point	Cover Type	Obstruction	Multipath
Center	Dwarf Scrub	Low	Low
North	Dwarf Scrub	Low	Low
South	Dwarf Scrub	Low	Low
East	Dwarf Scrub	Low	Low
West	Dwarf Scrub	Low	Low

Dominant cover: Dwarf Scrub · Source: USGS NLCD 2024 via MRLC · 30 m resolution · Industry standard mask guidance per NGS field procedures

Monday, Jul 13

Overcast

9 mph

GO

High 93°F

Low 58°F

Wind max 9 mph

Gusts 12 mph

Precip 0.00"

Sunrise 5:56 AM

Sunset 8:33 PM

Waning Crescent 2%

GNSS: Avg PDOP 0.99 · Best 4-hr window 6:00 AM – 10:00 AM (PDOP 0.94, 34 sats avg)

GNSS / DOP - HOURLY

	PDOP	SATS	RATING
6:00 AM	0.84	36 sv	IDEAL
7:00 AM	0.95	34 sv	IDEAL
8:00 AM	0.95	33 sv	IDEAL
9:00 AM	1.00	33 sv	GOOD
10:00 AM	1.01	29 sv	GOOD
11:00 AM	1.11	25 sv	GOOD
12:00 PM	1.03	26 sv	GOOD
1:00 PM	1.08	26 sv	GOOD
2:00 PM	0.93	32 sv	IDEAL
3:00 PM	0.97	33 sv	IDEAL
4:00 PM	0.98	33 sv	IDEAL
5:00 PM	0.99	33 sv	IDEAL
6:00 PM	1.02	30 sv	GOOD
7:00 PM	0.99	32 sv	IDEAL



GO - GOOD CONDITIONS

Partly cloudy | Wind: 9.4 mph (gusts 11.6 mph) | Precip: 0" | Temp: 58.2°–93.2°F



DRONE WIND ANALYSIS



GO Peak: 9.4/11.6 mph S

* Sunrise: 5:56 AM Sunset: 8:33 PM fly window clipped to daylight

Best flight window: 6:00 AM – 9:00 PM (15 hrs)

Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≈turbulence	DIR
5 AM	1/3 mph		NE
6 AM	1/2 mph		WNW
7 AM	1/2 mph		SW
8 AM	3/4 mph		S
9 AM	3/3 mph		SSE
10 AM	4/5 mph		S
11 AM	4/4 mph		SSE
12 PM	6/7 mph		S
1 PM	5/5 mph		S
2 PM	9/10 mph		S
3 PM	7/8 mph		S
4 PM	8/9 mph		SSW
5 PM	9/11 mph		SSW
6 PM	9/11 mph		SSW
7 PM	9/12 mph		S
8 PM	8/11 mph		S

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ▶ FREE TO FLY

Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107

→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY ↗](#)

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

FIELD PLANNING REPORT

JOB: SG-DQYW0D

SITE: Napa County, CA (38.3459, -122.3047)

DATE: Monday, July 13, 2026

REPORT ISSUED: Monday, July 13, 2026

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VERDICT: GO

PRIMARY REASONING: Excellent PDOP geometry (avg 0.99, best window 0.94) with 34 satellites available. Moderate ionospheric activity manageable with standard static protocols. Weather acceptable. No operational constraints.

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BEST SURVEY WINDOW: 6:00 AM – 10:00 AM LOCAL TIME

This 4-hour window delivers peak constellation geometry (PDOP 0.94) and optimal satellite count (34 sats). Morning timing avoids afternoon heating/refraction. Ionospheric delay will be present but manageable; recommend standard static occupation times (20–30 min per setup minimum). Window precedes overcast conditions solidifying.

=====

BEST 4-HOUR STATIC GNSS SESSION: 6:30 AM – 10:30 AM LOCAL TIME

Center the occupation within the high-geometry window. Use 15° elevation mask (confirmed against site terrain). Allow 25–30 minute sessions per point minimum given moderate Kp activity (4.7 peak). Expect ionospheric delay ~0.5–1.0 m; post-processing with dual-frequency will recover accuracy to <5 cm.

=====

DRONE OPERATIONS ASSESSMENT: APPROVED WITH CONDITIONS

Wind forecast 9 mph maximum is within normal UAS operating limits. Overcast conditions reduce solar panel efficiency if using solar-equipped platforms; standard battery capacity sufficient for typical mission duration. No airspace conflicts. Coordinate drone flights outside primary GNSS windows if possible to avoid line-of-sight interference; morning window (6:00–10:00 AM) should remain GNSS-priority.

=====

FIELD SAFETY WARNINGS

- No active wildfires in reporting range; all-clear for operations.
- GPS PRN 11 currently unhealthy (UNUSUFN); receiver firmware will auto-exclude. No manual intervention required.
- Peak AQI is 45 (Good); air quality acceptable.
- Overcast conditions typical for region; ensure crew hydration and sun protection during morning operations (high 93°F).
- Terrain elevation mask 13° recommended; orient antennas accordingly.

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SUMMARY

Monday, July 13, 2026 is a GO for GNSS survey operations. Constellation geometry is excellent (PDOP 0.99 average, best window 0.94 with 34 satellites) despite moderate geomagnetic activity (Kp 4.7 peak). Target the 6:00–10:00 AM window for optimal results; static sessions of 25–30 minutes will tolerate ionospheric delay and yield <5 cm post-processed accuracy. Weather is acceptable (overcast, low precipitation, winds <10 mph), drone operations are feasible, and site conditions (dwarf scrub, low multipath) support quality results. No critical satellite outages impact this survey date.

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COUNTY RECORDING OFFICES

RECORDER / REGISTER OF DEEDS

Napa County Recorder-Clerk

ASSESSOR / TAX OFFICE

Napa County Assessor

CORS STATIONS – WITHIN 30 MILES

QR / LINK	STATION ID	NAME / LOCATION	DISTANCE
 NGS ↗	P261	HUNTERHILLCN2004	14.1 mi
 NGS ↗	P198	PETALUMAIRC2004	17.5 mi
 NGS ↗	OHLN	OHLONE PARK	23.5 mi
 NGS ↗	P196	MEACHUMLFLCN2006	24.0 mi
 NGS ↗	CASR	SANTA ROSA CA	24.9 mi
 NGS ↗	P267	DIXONAVIATCN2005	26.3 mi

NGS BENCHMARKS – WITHIN 10 MILES

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
 Datasheet ↗	JT9010	"13"	19.100m / 62.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9018	"13 E"	17.970m / 59.0ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9009	"13"	17.900m / 58.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9017	"13 D"	17.890m / 58.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9034	"20 A"	20.730m / 68.0ft NAVD88	Order	0.3 mi
 Datasheet ↗	JT9012	"13"	19.060m / 62.5ft NAVD88	Order	0.4 mi
 Datasheet ↗	JT9043	"25 A"	18.710m / 61.4ft NAVD88	Order	0.4 mi
 Datasheet ↗	JT9016	"13 D"	18.200m / 59.7ft NAVD88	Order	0.5 mi

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
Datasheet ↗					

FEMA FLOOD ZONE

X – AREA OF MINIMAL FLOOD HAZARD

Panel: 06055C0510FF

Effective: Sep 29, 2010

[View FEMA Map ↗](#)



DATA COLLECTOR IMPORT – NGS BENCHMARKS

NGS BENCHMARK CONTROL POINTS – Copy / Paste to Data Collector

Datum: NAD83 | Vertical: NAVD88 | Source: NGS NDE API

PID	NAME	LAT	LON	ELEV_FT	ORDER
JT9010	13	38.348333	-122.302778	62.7	
JT9018	13 E	38.348333	-122.302778	59.0	
JT9009	13	38.346667	-122.301111	58.7	
JT9017	13 D	38.345000	-122.301111	58.7	
JT9034	20 A	38.345000	-122.309444	68.0	
JT9012	13	38.351667	-122.304444	62.5	
JT9043	25 A	38.340000	-122.302778	61.4	
JT9016	13 D	38.341667	-122.297778	59.7	

Download .txt File

USGS LAKES & RESERVOIRS – within 150 miles

STATION	DIST	LEVEL	FULL POOL	LINK
CLEAR LK A LAKEPORT CA #11450000	58.1 mi	5.05 ft	N/R	USGS ↗

CALIFORNIA RESERVOIRS (DWR/CDEC) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
Lake Del Valle	60.6 mi	696.94 ft (-9 ft)	706 ft	36K af	67%
Folsom Lake	66.5 mi	448.19 ft (-18 ft)	466 ft	787K af	81%
Lake Oroville	93.4 mi	861.2 ft (-39 ft)	900 ft	2,859K af	81%
New Melones Lake	100.1 mi	1,020.19 ft (-68 ft)	1,088 ft	1,664K af	69%
Black Butte Lake	101.2 mi	—	228 ft	74K af	54%

USBR RESERVOIRS (Bureau of Reclamation) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
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ARMY CORPS RESERVOIRS (USACE) – within 150 miles

RESERVOIR	DIST	POOL ELEV	FULL POOL	STORAGE	% CAP
Del Valle	64 mi	696.9 ft	695.01 ft	—	100%
Camanche	89 mi	230.3 ft	221.86 ft	—	104%
Englebright Lake-Pool	95 mi	525.1 ft	522.35 ft	—	101%
Farmington Dam-Pool	99 mi	124.9 ft	117.09 ft	—	107%
Black Butte-Pool	101 mi	457 ft	458.36 ft	—	100%
Don Pedro	137 mi	—	N/R	—	—

USGS GROUNDWATER WELLS – within 15 miles

STATION	DIST	WELL DEPTH	WATER LEVEL	DATE	AQUIFER
005N004W32A002M	7 mi	200 ft	87.71 ft BGS	2014-11-05	N100CACSTL
005N004W28R001M	7 mi	51 ft	41.90 ft BGS	1977-10-03	N100CACSTL
004N004W04C001M	8 mi	80 ft	11.20 ft BGS	1977-10-03	N100CACSTL
004N004W05B001M	8 mi	—	54.10 ft BGS	1977-10-03	N100CACSTL
005N004W31R001M	8.1 mi	295 ft	—	—	—
004N004W05D002M	8.3 mi	60 ft	14.60 ft BGS	1977-10-03	N100CACSTL
004N004W06B001M	8.3 mi	201 ft	21.86 ft BGS	2022-06-07	—
004N004W02L001M	8.8 mi	—	9.90 ft BGS	1977-10-02	N100CACSTL

○ AI AQUIFER INTELLIGENCE — CLAUDE HAIKU

The California Coastal Ranges Sedimentary and Volcanic aquifer (N100CACSTL) in this part of Napa County consists of fractured volcanic rocks and interbedded sedimentary deposits that formed during the Tertiary era as part of the Coast Ranges geologic system. Water moves slowly through fractures and permeable layers, making this a moderate-yield aquifer that responds to both seasonal recharge and longer-term climate cycles. At this location, water levels range from about 10 to 88 feet below ground surface, indicating the aquifer is accessible but requires drilling to moderate depths.

FORMATION

Name	California Coastal Ranges Sedimentary and Volcanic Aquifer
Age / Rock	Tertiary-age fractured volcanic rock and interbedded sedimentary deposits
Extent	2000–3500 sq mi (est. USGS/DWR)

DEPTH & RECHARGE

Depth Range	9–295 ft (1–30 floors)
Typical Depth	60–120 ft (6–12 floors)
Recharge Source	Precipitation on upland outcrops and hillslope seepage through fractured volcanic formations

Transit Time	20–50 years to 100 ft depth
WATER QUALITY	
pH	est. 7.0–7.8 (est. USGS/DWR)
Hardness	est. 150–350 mg/L CaCO3 (est. USGS/DWR)
TDS	est. 250–500 mg/L (est. USGS/DWR)
Notes	No sample data — typical for formation; volcanic terranes may locally elevate silica and minor metals
TREND	
Trend	Water levels show long-term decline of approximately 66 feet over 45 years (1977–2022), with steeper declines in deeper wells, indicating sustained drawdown in the aquifer system
Source	Site historical water-level measurements (1977–2022)
Seasonal Swing	est. 8–20 ft seasonal swing (est. USGS/DWR)

Source: USGS National Water Information System · Search all wells in this area → USGS NWIS Mapper ↗ · Water level data pending USGS API migration for CA state well numbers

LAND SUBSIDENCE & AQUIFER COMPACTION

✓ **LOW SUBSIDENCE RISK — No known critically overdrafted basin**
Source: CA DWR SGMA Bulletin 118 2024.1 · California only - full US pending server migration

🕒 **AI SUBSIDENCE ASSESSMENT — CLAUDE HAIKU**

The study area near Sonoma County (38.3459, -122.3047) is located in a region with well-drained soils and is underlain by the North Coast Coastal aquifer system (N100CACSTL), which has not experienced significant groundwater overdraft. The absence of extensive clay aquitards and the moderate groundwater extraction rates in this coastal terrain have prevented the compaction-driven subsidence observed in California's Central Valley and other critically overdrafted basins. For surveyors operating in this area, published benchmarks and monuments are expected to remain stable with minimal vertical drift, making traditional elevation control methods reliable without subsidence corrections.

SUBSIDENCE ASSESSMENT	
Susceptibility	Low
Primary Mechanism	None - consolidated terrain
Annual Rate	<1 mm/yr (est. USGS)
Historic Loss	None documented
Aquifer Compaction	None - aquifer not over-pumped
SGMA Status	Not applicable - not in SGMA basin
BENCHMARK IMPLICATIONS	
Monument Risk	Low - monuments stable
Elevation Validity	Good - no active subsidence
GNSS Recommendation	Standard benchmark verification adequate
SURVEY ACTION	
Required Action	No special action required
Dataset	CA DWR SGMA Bulletin 118 2024.1
Coverage	California critically overdrafted basins only - full US coverage pending

STRUCTURAL & ENGINEERING PARAMETERS

SEISMIC (ASCE 7-22)		WIND / SNOW / ICE / FROST (ASCE 7-22)	
Site Class	D (default)	Basic Wind Speed	92 mph
Risk Category	II	Ground Snow Load	5.8 psf
		Winter Wind (W2)	0.35
		Ice Thickness	None
		Frost Depth	12 in minimum
		Est. from IECC Zone 3C — AFI data unavailable	
		Verify with local building official before construction	
		Source: ASCE 7-22 Hazard Tool · gis.asce.org	

CLIMATE ZONE (IECC 2021): Zone 3C · Marine · Marine - mild year-round, coastal influence (SF, LA coast) County: Napa

🕒 **AI CLIMATE SUMMARY — RESIDENTIAL DESIGN**

This marine climate has mild, consistent temperatures year-round, so your primary envelope concern is managing solar gain and moisture rather than extreme heating or cooling—prioritize high-quality air sealing and vapor barriers on the ocean-facing side. Concentrate glazing on the equator-facing (south) elevations to capture winter sun, and minimize west-facing glass or use external shading to prevent afternoon overheating, while keeping the windward (typically west/northwest) side tight against the 92 mph coastal winds. The single most effective passive strategy for this zone is natural cross-ventilation through operable windows positioned to catch the reliable coastal breezes, allowing you to avoid mechanical cooling for much of the year.

🕒 **AI STRUCTURAL ASSESSMENT — CLAUDE HAIKU**

This coastal marine site in IECC Zone 3C presents moderate wind exposure and light snow loading with well-drained gravelly soils anticipated to support conventional foundations. Seismic parameters are unavailable; a geotechnical investigation may warrant confirmation of Site Class D and assessment of any liquefaction screening needs. Frost depth of approximately 12 inches should be verified with the local building official and may influence footing design.

FOUNDATION	
Type	Conventional spread footing anticipated

Basis Well-drained gravelly loam soils suggest adequate bearing capacity for typical residential or light commercial loads. Consult a licensed geotechnical engineer to confirm bearing capacity and settlement characteristics, and to verify frost depth requirements.

STRUCTURAL IMPLICATIONS

- Seismic** Seismic design parameters are currently unavailable (null values). A liquefaction screening is suggested for Site Class D soils in this region. Consult a structural engineer to confirm whether special moment frames or other seismic detailing may be warranted pending SDC determination.
- Wind Exposure** Exposure C (open terrain) is indicated for ASCE 7-22 Risk Category II with 92 mph 3-second gust wind speed.
- Wind** Standard wind load design anticipated per ASCE 7-22. No hurricane exposure anticipated at this latitude and coastal proximity; marine wind loading effects may warrant consideration —verify site conditions with local wind data.
- Snow** Light snow load of 5.8 psf anticipated with winter wind factor W2=0.35. Standard roof design likely adequate; consult engineer for roof pitch and exposure effects.

INVESTIGATIONS & INSPECTIONS

- Geotech Required** Yes—geotechnical investigation recommended to confirm Site Class D, verify bearing capacity, assess liquefaction screening need, and confirm frost depth for local jurisdiction.
- Liquefaction Screen** Screening recommended for Site Class D soils. Groundwater depth and soil saturation should be evaluated; consult a geotechnical engineer for site-specific assessment.
- Special Inspections** Not anticipated unless seismic parameters warrant SDC assignment. Confirm with structural engineer once seismic design parameters are obtained.
- Code Edition** ASCE 7-22 / IBC 2021 (planning reference only)
- Site Class Note** Site Class D assumed for planning purposes. Geotechnical investigation strongly recommended to confirm soil classification, bearing capacity, and subsurface conditions before final design.

SOILS — USDA SOIL SURVEY

MAP UNIT: Coombs gravelly loam, 2 to 5 percent slopes	SERIES: Coombs
DRAINAGE CLASS: Well drained	HYDRO GROUP: C
SOIL ORDER: Alfisols	SUBORDER: Xeralfs

AI SOIL INTELLIGENCE — CLAUDE HAIKU

Coombs gravelly loam is a well-drained Alfisol (Xeralf) formed in alluvial and colluvial materials on gentle slopes in the coastal ranges and valleys of California. The soil profile features a gravelly loam surface layer underlain by a clay loam B horizon with moderate structure, providing moderate bearing capacity suitable for most light to moderate construction. Surveyors and developers should expect good trafficability, moderate excavation resistance from gravel content, and stable foundation conditions on these 2–5% slopes, though subsurface clay warrants moisture control during fill placement.

Alluvial and colluvial gravels and loams on low hills and fan remnants; mixed oak and grass vegetation.

DRAINAGE

Class	Well drained
Seasonal Saturation	None within 60 inches
Flooding Risk	Minimal on 2–5% slopes
Ponding Risk	None

ENGINEERING

Bearing Capacity	3,000–4,000 psf (shallow foundations)
Rating	Good
Shrink-Swell	Low to Moderate — clay loam substratum with moderate expansion potential
Frost Action	Low
Cut Suitability	Good (15–35% gravel content aids drainage and stability)
Fill Suitability	Fair (clay loam substratum — moisture control required during placement)

SEPTIC

Suitability	Conditional
Notes	Perc test required — clay loam B horizon may limit infiltration rates; site-specific evaluation recommended

HAZARDS

Excavation	Moderate — 15–35% gravel content increases equipment wear
Expansive Clay	Low anticipated
Erosion	Low on 2–5% slopes; moderate gully risk if drainage is disrupted
Other	None

Data sources: USGS National Water Information System · waterdata.usgs.gov · California DWR / CDEC · cdec.water.ca.gov · USBR RISE · data.usbr.gov · USACE CWMS · water.usace.army.mil · USGS NWIS Groundwater · waterservices.usgs.gov · USDA Web Soil Survey · sdmdataaccess.nrcs.usda.gov · CA DWR SGMA Bulletin 118 · gis.water.ca.gov · ASCE 7-22 Hazard Tool · gis.asce.org · USGS Design Maps · earthquake.usgs.gov · IECC 2021 Climate Zones · DOE Building America · Real-time data subject to revision

Water levels current as of report run: Jul 13, 2026, 2:03 AM UTC

* Live pool elevation not reported by this reservoir — full pool design elevation shown

OIL & GAS WELLS California — within 10mi radius

8 total
0 active
7 plugged
 By status:
 7 Plugged
 1 Idle

By type: 8 DH

Lease / Well	Status	Type	Operator	Depth	Dist
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	7.2 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	7.2 mi
McKenzie-Hazard Syndicate	Plugged	DH	McKenzie-Hazard Syndicate	—	7.4 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	8.6 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	8.8 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	9.5 mi
Capell Oil Corp.	Idle	DH	Capell Oil Corp.	—	9.7 mi
Cordelia Oil & Gas Co.	Plugged	DH	Cordelia Oil & Gas Co.	—	12 mi

AI Assessment — Well Proximity

Based on the well data for this site, oil and gas well proximity is not a planning concern for this project. The nearest well is a plugged Mobil Oil well located 7.2 miles away, which is well beyond all actionable distance thresholds. There are no active producers, injection wells, or idle wells within relevant setback distances that would require further review.

Source: California public well records. Active = currently producing or permitted. Plugged = abandoned/decommissioned. Data updated nightly.

200 **10** **108**
total in search area within 5 mi within 25 mi

Site Name	Status	Commodities	Distance
Roderick Sand and Gravel Pit	Occurrence	Sand/Gravel	1.6 mi
Basalt Rock Co	Producer	Clay	2 mi
Cicero Pumice Quarry	Occurrence	PUM	2.9 mi
Mount George King Pumice Deposit.	Occurrence	PUM	2.9 mi
Basalt Rock Co.	Occurrence	DIT	3.1 mi
Mt George Pumice Deposit	Producer	PUM	3.1 mi
Wilson Pumice Quarry	Occurrence	PUM	3.7 mi
Unknown	Occurrence	Mercury	4.4 mi
Silva Pumice Quarry	Occurrence	PUM	4.7 mi
Silva Pumice Quarry	Producer	PUM	5 mi
Walker Pumice Quarry	Occurrence	PUM	5.1 mi
Walker Pumice Quarry	Producer	PUM	5.1 mi
Mountain	Occurrence	Mercury	5.4 mi
Unknown	Occurrence	Mercury	7.6 mi
Unknown	Occurrence	Stone/Crushed	7.8 mi
Unnamed Location	Past Producer	Sand/Gravel	8.5 mi
Unnamed Location	Past Producer	Sand/Gravel	8.5 mi
Unnamed Prospect	Occurrence	Chromium	8.8 mi
Unnamed Location	Occurrence	Chromium	9 mi
Unknown	Occurrence	Chromium	9.1 mi
Bella Oak	Past Producer	Mercury	9.4 mi
Unknown	Occurrence	MG	9.7 mi
Serres Pit	Past Producer	Sand/Gravel	10.4 mi
Unnamed Location	Unknown	Sand/Gravel	10.5 mi
Conn Valley	Occurrence	Manganese	10.6 mi

Showing 25 of 200 deposits. Full dataset via USGS MRDS.

AI Assessment — Mining Context

Three active mineral producers operate within 5 miles of the site. Basalt Rock Co (2 mi, clay) and Mt George Pumice Deposit (3.1 mi, pumice) and Silva Pumice Quarry (5 mi, pumice) will generate heavy truck traffic on access roads during operating hours, dust emissions from extraction and processing, and potential noise from blasting or crushing operations. No inactive deposits warrant Phase 1 assessment consideration given they all exceed 1 mile proximity.

Source: USGS Mineral Resources Data System (MRDS). Status: Producer = active/recent; Past Producer = historical; Prospect = explored not mined; Occurrence = mineralization noted.

ENDANGERED & THREATENED SPECIES - USFWS Critical Habitat within 25mi

6 **6** **12**
Endangered Threatened Total w/ Critical Habitat

6 Endangered species with designated critical habitat. ESA Section 7 consultation may be required for federal nexus projects.

COMMON NAME	SCIENTIFIC NAME	STATUS
California tiger Salamander	<i>Ambystoma californiense</i>	Endangered
Conservancy fairy shrimp	<i>Branchinecta conservatio</i>	Endangered
Contra Costa goldfields	<i>Lasthenia conjugens</i>	Endangered
Soft bird's-beak	<i>Cordylanthus mollis</i> ssp. <i>mollis</i>	Endangered
Suisun thistle	<i>Cirsium hydrophilum</i> var. <i>hydrophilum</i>	Endangered
Vernal pool tadpole shrimp	<i>Lepidurus packardii</i>	Endangered
Alameda whipsnake (=striped racer)	<i>Masticophis lateralis euryxanthus</i>	Threatened
California red-legged frog	<i>Rana draytonii</i>	Threatened
Delta smelt	<i>Hypomesus transpacificus</i>	Threatened
Northern spotted owl	<i>Strix occidentalis caurina</i>	Threatened
Vernal pool fairy shrimp	<i>Branchinecta lynchi</i>	Threatened
western snowy plover	<i>Charadrius nivosus nivosus</i>	Threatened

AI SPECIES & ESA ASSESSMENT - CLAUDE HAIKU

ESA Implications Assessment – 38.3459, -122.3047 **ESA Section 7 Consultation Triggers:** This location's proximity to critical habitat for 12 listed species will trigger mandatory Section 7 consultation if the proposed project involves federal permitting, funding, or authorization. The presence of six endangered species and six threatened species significantly elevates compliance requirements and timeline considerations. **Planning-Relevant Priority Species:** California tiger salamander, vernal pool tadpole shrimp, and vernal pool fairy shrimp warrant immediate focus due to their dependence on seasonal wetland habitats likely present in this region; California red-legged frog and Alameda whipsnake are secondary priorities given their broader habitat requirements. The suite of vernal pool-dependent species suggests that site hydrology and seasonal ponding areas will be the primary constraints on project feasibility. **Recommended Next Steps:** Conduct a Phase 1 biological resources survey to delineate potential habitat (particularly seasonal wetlands, ephemeral pools, and grasslands), then initiate early coordination with USFWS and CDFG to understand project-specific consultation pathways. Integrate findings into project design alternatives to avoid or minimize impacts, which will streamline permitting and reduce project delays. *For planning purposes only. Consult a licensed environmental biologist for formal ESA compliance.*

Source: USFWS Critical Habitat Final Designated Areas - FWS Open Data - Critical habitat does not equal species range

NATIVE PLANTS - iNaturalist Research-Grade within 10mi

57 **5051** **2026-07-04**
Unique Species Total Observations Most Recent

COMMON NAME	SCIENTIFIC NAME	LAST OBS
American black nightshade	<i>Solanum americanum</i>	2026-05-29
Bearded Clover	<i>Trifolium barbigerum</i>	2026-05-15
bird's foot cliffbrake	<i>Pellaea mucronata</i>	2026-04-28
Blow Wives	<i>Achyrachaena mollis</i>	2026-05-20
blue oak	<i>Quercus douglasii</i>	2026-05-09
Bolander's Phacelia	<i>Phacelia bolanderi</i>	2026-04-27
box elder	<i>Acer negundo</i>	2026-05-18
Braun's Giant Horsetail	<i>Equisetum telmateia braunii</i>	2026-06-22
California bay	<i>Umbellularia californica</i>	2026-06-24
California beepplant	<i>Scrophularia californica</i>	2026-05-20
California black oak	<i>Quercus kelloggii</i>	2026-05-15
California buckeye	<i>Aesculus californica</i>	2026-06-02
California Dutchman's Pipe	<i>Aristolochia californica</i>	2026-05-26
California milkwort	<i>Rhinotropis californica</i>	2026-05-15
California mugwort	<i>Artemisia douglasiana</i>	2026-06-02
California poppy	<i>Eschscholzia californica</i>	2026-06-24
California Wild Rose	<i>Rosa californica</i>	2026-05-09
Canyon liveforever	<i>Dudleya cymosa</i>	2026-05-30
-	<i>Castilleja densiflora densiflora</i>	2026-05-09
clay mariposa lily	<i>Calochortus argillosus</i>	2026-05-15
coast live oak	<i>Quercus agrifolia</i>	2026-05-13
coastal woodfern	<i>Dryopteris arguta</i>	2026-05-20
common selfheal	<i>Prunella vulgaris</i>	2026-06-22
Diogenes' lantern	<i>Calochortus amabilis</i>	2026-05-15
distant phacelia	<i>Phacelia distans</i>	2026-05-15
-	<i>Downingia concolor concolor</i>	2026-05-15
Dwarf Brodiaea	<i>Brodiaea terrestris</i>	2026-05-15
Elegant Clarkia	<i>Clarkia unguiculata</i>	2026-05-16

COMMON NAME	SCIENTIFIC NAME	LAST OBS
goldback fern	<i>Pentagramma triangularis</i>	2026-05-20
Ithuriel's Spear	<i>Triteleia laxa</i>	2026-06-15
maroonspot calicoflower	<i>Downingia concolor</i>	2026-05-15
narrowleaf milkweed	<i>Asclepias fascicularis</i>	2026-06-25
narrowleaf mule-ears	<i>Wyethia angustifolia</i>	2026-05-15
Narrowleaf Willow	<i>Salix exigua</i>	2026-06-24
needleleaf navarretia	<i>Navarretia intertexta</i>	2026-05-15
northern California black walnut	<i>Juglans hindsii</i>	2026-06-24
orange bush monkeyflower	<i>Diplacus aurantiacus</i>	2026-05-29
Oregon Ash	<i>Fraxinus latifolia</i>	2026-06-02
Pacific Bleeding Heart	<i>Dicentra formosa</i>	2026-05-04
Pacific madrone	<i>Arbutus menziesii</i>	2026-07-04
Pacific poison oak	<i>Toxicodendron diversilobum</i>	2026-06-02
pineapple-weed	<i>Matricaria discoidea</i>	2026-05-20
Redwood Sorrel	<i>Oxalis smalliana</i>	2026-05-28
small-headed clover	<i>Trifolium microcephalum</i>	2026-05-15
Superb Mariposa Lily	<i>Calochortus superbus</i>	2026-06-11
thimble clover	<i>Trifolium microdon</i>	2026-05-15
tomcat clover	<i>Trifolium willdenovii</i>	2026-05-15
Toyon	<i>Heteromeles arbutifolia</i>	2026-06-19
trailing blackberry	<i>Rubus ursinus</i>	2026-06-24
Tree Clover	<i>Trifolium ciliolatum</i>	2026-05-15
valley oak	<i>Quercus lobata</i>	2026-06-24
wavy-leafed soap plant	<i>Chlorogalum pomeridianum</i>	2026-05-20
western sycamore	<i>Platanus racemosa</i>	2026-05-17
white brodiaea	<i>Triteleia hyacinthina</i>	2026-05-15
white-tipped clover	<i>Trifolium variegatum</i>	2026-05-15
Winecup Clarkia	<i>Clarkia purpurea</i>	2026-05-15
Yellow Mariposa Lily	<i>Calochortus luteus</i>	2026-06-11

AI PLANT COMMUNITY SUMMARY - CLAUDE HAIKU

Ecological Summary This site in Sonoma County supports a **mixed oak woodland and chaparral community**, dominated by native oaks (blue oak, California black oak) with scattered California bay, buckeye, and dense understory of forbs and shrubs typical of interior Coast Ranges. The presence of moisture-loving species like Braun's Giant Horsetail and clay mariposa lily alongside drought-tolerant chaparral plants (Canyon liveforever, Blow Wives) indicates **seasonally variable soil moisture—wet in winter/spring, dry by summer**—typical of clay or clay-loam soils common to this region. For grading and drainage planning, expect **slow water infiltration and seasonal perching of water** during wet months, with potential for erosion and runoff; summer conditions will be severely dry. Native plant establishment and erosion control should account for this pronounced seasonal moisture cycle. Consult a licensed botanist for site-specific vegetation assessment.

Source: iNaturalist research-grade observations - Native species filter applied

FIELD HAZARDS - Venomous & Toxic Species within 25mi

635	154	91	1439
Venomous Snakes	Venomous Spiders	Stinging Insects	Toxic Plants
VENOMOUS SNAKES			
SPECIES		OBSERVATIONS	
Crotalus oreganus oreganus		635 obs	
VENOMOUS SPIDERS			
SPECIES		OBSERVATIONS	
Latrodectus hesperus		154 obs	
STINGING INSECTS			
SPECIES		OBSERVATIONS	
Dasymutilla aureola		91 obs	
Dasymutilla californica		91 obs	
Dasymutilla sackenii		91 obs	
TOXIC PLANTS			

SPECIES	OBSERVATIONS
<i>Toxicodendron diversilobum</i>	1100 obs
<i>Datura wrightii</i>	57 obs
<i>Datura stramonium</i>	57 obs
<i>Datura ferox</i>	57 obs
<i>Conium maculatum</i>	199 obs
<i>Urtica gracilis</i>	83 obs
<i>Urtica gracilis holosericea</i>	83 obs

AI FIELD SAFETY BRIEFING - CLAUDE HAIKU

Field Safety Brief – Northern California Survey Area **Highest Priority Hazards:** Your greatest risks are toxic plants—particularly poison oak (*Toxicodendron diversilobum*), which is extremely common in this region and causes severe skin reactions on contact. Northern Pacific rattlesnakes (*Crotalus oreganus oreganus*) are the second concern; they’re present but generally avoid humans. Black widow spiders and stinging insects pose lower but real risk. **Practical Avoidance:** Wear long pants, long sleeves, and closed-toe boots; learn to identify poison oak (three leaflets, waxy coating) and avoid touching it. Stay on cleared survey lines and avoid reaching into brush. Watch where you place hands and feet, particularly in rocky areas and near vegetation. Carry a first aid kit and know the location of nearest medical support. **Seasonal Note:** Rattlesnakes are most active April–October; toxic plants are hazardous year-round but more visible in growing season. Stinging insects peak in late summer. Always verify current conditions locally before fieldwork.

Source: iNaturalist research-grade observations - Always verify locally before fieldwork

FCC MICROWAVE PATHS — Licensed beam paths crossing within 1320ft of site

3	545ft	3
Beam Paths	Closest Beam	Fixed P2P

TX CALL	RX CALL	TYPE	PATH MI	CLOSEST
WBO61	WBO59	Fixed Point-to-Point	—	545ft
KYH37	WQAT627	Fixed Point-to-Point	—	927ft
WRCR329	WQAT627	Fixed Point-to-Point	—	1289ft

Source: FCC ULS microwave license database — PostGIS spatial query — Paths within 1320ft (0.25mi) of site — Towers may be 50+ miles away — FCC Part 101 licensed paths

FAA AIRWAYS — ATS Routes crossing site area

26	13	2	11
Total Airways	Jet Routes	Victor Airways	Other Routes

AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
J1	Jet Route	High	18,000 ft	—	303.8°	121.4°
J110	Jet Route	High	18,000 ft	—	137.9°	318.2°
J126	Jet Route	High	18,000 ft	—	325.5°	144.0°
J143	Jet Route	High	18,000 ft	—	325.0°	145.8°
J3	Jet Route	High	18,000 ft	—	342.8°	161.8°
J32	Jet Route	High	18,000 ft	—	19.2°	199.6°
J501	Jet Route	High	18,000 ft	—	317.1°	135.4°
J58	Jet Route	High	18,000 ft	—	65.3°	245.9°
J65	Jet Route	High	18,000 ft	—	304.1°	121.1°
J80	Jet Route	High	18,000 ft	—	65.3°	245.9°
J84	Jet Route	High	18,000 ft	—	50.8°	231.0°
J88	Jet Route	High	18,000 ft	—	308.0°	127.3°
J94	Jet Route	High	18,000 ft	—	65.3°	245.9°
Q1	Q-Route	High	—	—	334.5°	154.1°
Q120	Q-Route	High	—	—	38.1°	219.8°
Q122	Q-Route	High	—	—	37.8°	219.6°
Q124	Q-Route	High	—	—	37.8°	219.6°
Q138	Q-Route	High	—	—	47.7°	229.0°
Q3	Q-Route	High	—	—	164.0°	344.6°
Q5	Q-Route	High	—	—	161.4°	341.7°
T257	T-Route	Low	—	—	321.1°	140.5°
T259	T-Route	Low	—	—	41.6°	221.8°
T261	T-Route	Low	—	—	31.4°	211.7°
T263	T-Route	Low	—	—	308.8°	128.2°
V107	Victor Airway	Low	7,000 ft	—	295.4°	114.2°

AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
V108	Victor Airway	Low	4,500 ft	—	117.7°	296.8°

Source: FAA ATS Route FeatureServer — Airways shown cross the site bounding area and may not pass directly overhead — MEA = Minimum En Route Altitude

RAILROAD PROXIMITY — FRA North American Rail Network

8
Rail Lines

OWNER / OPERATOR	TYPE	TRACKS	PASSENGER	STRACNET	SUBDIVISION
CFNR	Other	1	No	No	—
CFNR	Yard	1	No	No	—
CFNR	Mainline	1	No	No	VALLEJO
SMRT	Mainline	1	No	No	BRAZOS JCT.
NVRR	Mainline	1	No	No	MARTINEZ
USG	Other	1	No	No	—
CFNR	S	1	No	No	—
NVRR	Yard	1	No	No	—

Source: FRA North American Rail Network (NARN) — Updated July 2025 — STRACNET = Strategic Rail Corridor Network (DOD)

WILDLIFE & NATURAL HISTORY - iNaturalist within 25mi

200 90 65 5 14 6
Total Observations Total Species Birds Mammals Reptiles Amphibians

BIRDS (65 species)

COMMON NAME	Scientific Name	OBS
Wild Turkey	<i>Meleagris gallopavo</i>	10
Snowy Egret	<i>Egretta thula</i>	7
California Quail	<i>Callipepla californica</i>	5
Western Bluebird	<i>Sialia mexicana</i>	5
California Towhee	<i>Melozone crissalis</i>	5
Great Egret	<i>Ardea alba</i>	4
Red-tailed Hawk	<i>Buteo jamaicensis</i>	4
Black Phoebe	<i>Sayornis nigricans</i>	4
Cooper's Hawk	<i>Astur cooperii</i>	3
Ash-throated Flycatcher	<i>Myiarchus cinerascens</i>	3
Western Screech-Owl	<i>Megascops kennicottii</i>	3
Turkey Vulture	<i>Cathartes aura</i>	3
Mallard	<i>Anas platyrhynchos</i>	3
Bushtit	<i>Psaltiriparus minimus</i>	3
Canada Goose	<i>Branta canadensis</i>	3

+ 50 additional species

MAMMALS (5 species)

COMMON NAME	Scientific Name	OBS
Black-tailed Jackrabbit	<i>Lepus californicus</i>	4
Coyote	<i>Canis latrans</i>	2
Mule Deer	<i>Odocoileus hemionus</i>	2
Eastern Fox Squirrel	<i>Sciurus niger</i>	1
Striped Skunk	<i>Mephitis mephitis</i>	1

REPTILES (14 species)

COMMON NAME	Scientific Name	OBS
Western Fence Lizard	<i>Sceloporus occidentalis</i>	10
Pacific Gopher Snake	<i>Pituophis catenifer catenifer</i>	7
Northern Pacific Rattlesnake	<i>Crotalus oreganus oreganus</i>	6
California King Snake	<i>Lampropeltis californiae</i>	3
Pond Slider	<i>Trachemys scripta</i>	2
Western Skink	<i>Plestiodon skiltonianus</i>	2

COMMON NAME	Scientific Name	OBS
Ring-necked Snake	<i>Diadophis punctatus</i>	1
Common Garter Snake	<i>Thamnophis sirtalis</i>	1
Western Yellow-bellied Racer	<i>Coluber constrictor mormon</i>	1
Pacific Ringneck Snake	<i>Diadophis punctatus amabilis</i>	1
Northwestern Pond Turtle	<i>Actinemys marmorata</i>	1
Common Sharp-tailed Snake	<i>Contia tenuis</i>	1
North American Racer	<i>Coluber constrictor</i>	1
Southern Alligator Lizard	<i>Elgaria multicarinata</i>	1

AMPHIBIANS (6 species)

COMMON NAME	Scientific Name	OBS
Pacific chorus frog	<i>Pseudacris regilla</i>	5
American Bullfrog	<i>Lithobates catesbeianus</i>	3
Foothill Yellow-legged Frog	<i>Rana boylei</i>	3
Western Toad	<i>Anaxyrus boreas</i>	2
Rough-skinned Newt	<i>Taricha granulosa</i>	1
California Giant Salamander	<i>Dicamptodon ensatus</i>	1

AI WILDLIFE & ECOLOGICAL ASSESSMENT - CLAUDE HAIKU

Ecological Character Assessment This site in Sonoma County, California supports a moderately diverse wildlife community characteristic of oak woodland and grassland habitats with adjacent riparian or wetland features, as evidenced by the co-occurrence of upland species (Wild Turkey, California Quail, mule deer) and wetland-dependent species (egrets, pond turtles, chorus frogs). Several species of regulatory concern are present, including Foothill Yellow-legged Frog and California Giant Salamander, which indicate the presence of perennial or seasonal streams with high water quality and intact riparian vegetation; Northwestern Pond Turtle and Rough-skinned Newt also suggest aquatic or semi-aquatic habitat connectivity. The species composition reflects a landscape with moderate habitat heterogeneity and relatively low to moderate human disturbance, though the presence of non-native species (Eastern Fox Squirrel, American Bullfrog, Pond Slider) indicates some degree of past land use alteration. Any site development should incorporate biological surveys to identify sensitive habitats, particularly aquatic and riparian areas, and implement measures to maintain wildlife corridors and protect sensitive species breeding and rearing habitat. **For planning purposes only. Consult a qualified biologist for site-specific biological assessments.**

Source: iNaturalist research-grade observations — Data reflects community science observations and may not represent complete species inventory

* PASSIVE + ACTIVE SOLAR PLANNING — Lat 38.35°

1 · AVERAGE SUN HOURS

Period	Peak Hrs/Day	Notes
Annual average	6.6 hrs	Overall solar resource
Summer average	9 hrs	June–August
Winter average	3.9 hrs	December–February
Worst month (Dec)	3.2 hrs	Size battery storage to this

2 · IDEAL SOLAR ARRAY PITCH

Optimal tilt = 38.3° (latitude) → 9/12 pitch

Pitch	Angle	% of Optimal	Notes
4/12	18.4°	97%	good
5/12	22.6°	97.6%	good
6/12	26.6°	98.2%	↪ excellent
7/12	30.3°	98.8%	↪ excellent
8/12	33.7°	99.3%	↪ excellent
9/12	36.9°	99.8%	↪ annual optimal
10/12	39.8°	99.8%	↪ excellent
12/12	45°	99%	↪ excellent

Going from 5/12 to 6/12 at framing costs ~\$800–1,200 one time. On a 10kW system that's +723 kWh/yr (~\$108 value/yr). Over 25yr panel life: ~\$2710 total gain.

3 · SOLAR HEAT GAIN — BTU/HR PER SQ FT

Use these values as inputs for eQUEST, EnergyPlus, Manual J, or HERS rating

Surface	Winter	Summer	Design Note
South glass (SHGC 0.4)	51	31	Minimize south glass
South glass (SHGC 0.6)	76	46	Clear/uncoated glass
North glass (any)	10	13	Negligible — diffuse only
East / West glass	27	104	△ Minimize — afternoon load
Roof — dark (absorb 0.95)	236	301	Highest load surface
Roof — light (absorb 0.35)	87	111	Cool roof — significant reduction
Roof — metal (absorb 0.25)	62	79	Best roof option for heat
South wall — dark	94	69	
South wall — light	31	23	Light color = major savings
Hardscape — asphalt	181	301	△ Heat island — avoid near structure
Hardscape — concrete	124	206	Also reflects onto adjacent walls
Hardscape — light pave	67	111	Recommended within 50ft
Pool surface	23	38	Low absorb + evaporative cooling

Concrete reflection onto south wall: ~4 BTU/hr-ft² additional load per 100sf of adjacent hardscape.

Pool specular reflection: angle-dependent — east/west pools can cause morning/afternoon glare into glass.

4 • ROOF PITCH SOLAR LOAD — BTU/HR PER 100 SQ FT (SUMMER)

Pitch	Dark Roof	Light Roof	Reduction vs Flat
0/12	301 BTU	111 BTU	baseline (worst)
4/12	241 BTU	89 BTU	-20% dark / -20% light
6/12	196 BTU	72 BTU	-35% dark / -35% light
8/12	166 BTU	61 BTU	-45% dark / -45% light
12/12	135 BTU	50 BTU	-55% dark / -55% light

Scale: multiply BTU × (your roof sf ÷ 100). Pitch + light color = biggest single cooling move.

5 • THERMAL TIME LAG — WALL ASSEMBLY

Peak outdoor temp ~15:00. Heat arrives at interior = peak + lag hours.

Assembly	Lag	Peak Interior Load	Note
2x4 frame (R-15)	0.6 hrs	~15:36 PM	R-value only — no mass benefit
2x6 frame (R-23)	0.9 hrs	~15:54 PM	R-value only — no mass benefit
2x8 frame (R-30)	1.2 hrs	~16:12 PM	R-value only — no mass benefit
2x10 frame (R-37)	1.5 hrs	~16:30 PM	good
2x12 frame (R-44)	1.8 hrs	~16:48 PM	good
ICF 6" core	7 hrs	~22:00 PM	✓ shifts past sunset — natural vent window
8" CMU block	6 hrs	~21:00 PM	✓ shifts past sunset — natural vent window
12" concrete	8.5 hrs	~23:30 PM	✓ shifts past sunset — natural vent window

In 3C Marine: ICF or CMU shifts peak load past sunset. Open windows, flush heat naturally. AC load drops significantly.

6 • TREE SHADOW PLANNING — TREE DUE SOUTH, 100FT FROM STRUCTURE

Tree Ht	Shadow Reaches	Summer hrs/day	Equinox hrs/day	Winter hrs/day	WUI note
10ft	All year	1.1	1	1.2	OK
20ft	All year	2.1	1.9	2.4	OK
30ft	All year	3.1	2.9	3.7	Zone 2 check
40ft	All year	4	3.8	5.2	△ verify
50ft	All year	4.8	4.6	7.3	△ verify
60ft	All year	5.6	5.5	9.3	✗ Zone 1

Deciduous tree: winter solar penetration ~85% (leaves off) — best of both worlds in hot-dry climates.

Evergreen tree: winter penetration ~15% — good windbreak, poor solar strategy south of structure.

△ WUI zones: tree placement must be coordinated with defensible space requirements. Verify with local fire authority — WUI boundaries vary by jurisdiction.

7 • ATTIC VENTING — PER 100 SQ FT OF ROOF AREA

Standard	Unbalanced (1/150)	Balanced† (1/300)	Notes
Code minimum	8 sq in	4 sq in	Moisture protection only
Thermal comfort	24 sq in	12 sq in	Keeps attic within 20°F of outdoor
WUI mesh 1/16"	60 sq in	30 sq in	2.5× compensation — mesh clogs
WUI intumescent	24 sq in	12 sq in	Self-closing — no compensation needed

Pitch factor: 0/12×1.0 · 4/12×0.80 · 6/12×0.65 · 8/12×0.55 · 12/12×0.45

Color factor: Dark×1.0 · Medium×0.75 · Light×0.55 · Metal×0.45

Scale: multiply sq in × (your roof sf ÷ 100) × pitch factor × color factor

†**Balanced** = ≥40% NFA within top third of roof height (ridge) + ≥40% within bottom third (soffit)

WUI conflict: Soffit vents are ember collection points. In high WUI zones — eliminate soffit vents, use ridge-only (unbalanced/1/150). In low WUI — balanced system preferred.

△ Verify WUI zone with local fire authority — boundaries vary significantly by jurisdiction.

🔗 AI SOLAR & PASSIVE DESIGN SUMMARY — HOMEOWNER GUIDE

Your Home's Solar & Energy Plan Think of your location like a solar sweet spot—you're getting about 6.6 hours of useful sunlight per day on average, which is solid for California, meaning solar panels here will work harder in summer (9 hours) but take it easier in winter (closer to 4). Your roof's pitch and color are actually your biggest levers for comfort and savings: a dark roof soaks up about 300 units of heat per hour per square foot in summer, while a light-colored one cuts that nearly in half, and your roof's natural angle of about 9 inches of rise per 12 inches of run is almost perfect for catching the sun year-round. Here's a critical one—your west-facing windows are heat traps in the afternoon, taking in twice as much summer warmth as your south windows, so those need shade, blinds, or strategic planting more than any other direction. A mature tree planted on your south side is genuinely smart here: it'll shade your home for about 4 hours in summer when you need it most, yet drop its leaves in winter to let the warming sun through during those cooler months. Thermal mass—think of it like your home's heat sponge—means that thicker, heavier walls (like ICF concrete) can absorb warmth during the day and release it slowly at night, keeping temperatures steadier than a standard wood-frame house would. These numbers are planning estimates—your architect or energy modeler can plug them directly into their software for a full analysis.

Planning estimates only — clear-sky model, standard assumptions. Values formatted for eQUEST / EnergyPlus / Manual J input. Verify with licensed architect, energy modeler, or solar installer before specifying.



Technical Brief — for architects, engineers, and surveyors

SITE INTELLIGENCE BRIEFING 38.34586°N, 122.30466°W | Napa County, California

Structural Design Drivers

The Coombs gravelly loam substrate and IECC Zone 3C marine climate classify this site as a moderate-load environment where seismic design does not emerge as the controlling parameter based on available data. No snow load (Snow: N/A) and no recorded wind data preclude wind or snow as design drivers. The AFI value of 0°F-days indicates negligible freezing-thaw cycling, which simplifies seasonal structural movement assumptions. **Structural design will be governed by gravity and dead load rather than climatic extremes or seismic amplification**, allowing consideration of conventional foundation systems without the material premium typical of higher-risk seismic zones or freeze-thaw environments.

Geotechnical & Foundation Considerations

The Coombs gravelly loam exhibits well-drained behavior within Hydrologic Group C, which typically indicates moderate infiltration and runoff characteristics suitable for conventional spread footing or shallow foundation systems. No frost depth data (Frost depth: N/A) is provided, but the AFI threshold of 0°F-days renders deep frost protection unnecessary—footings do not require seasonal adjustment for heave prevention. The gravelly loam composition suggests adequate bearing capacity and low settlement risk; however, the notation "texture: ?" indicates incomplete soil profile characterization in the database. **Subsurface exploration (SPT or CPT) is required to confirm bearing capacity, confirm the absence of lenses or seams, and establish design allowable soil pressures**—do not assume uniform performance across the site based on surficial mapping alone.

Hazard Assessment

No Quaternary faults appear within the search radius (Faults: N/A), and no subsidence basin designation has been recorded (Subsidence: N/A), eliminating fault rupture and ground collapse as design drivers. Wildfire data are not provided. The seismic parameter dataset is marked N/A, which indicates either that this location falls outside the California Geological Survey's detailed seismic hazard mapping, or that the query did not return results; clarify this with the state seismic hazard program or local building official to confirm design code applicability (ASCE 41, CBC provisions). **Until confirmed otherwise, assume standard California Building Code seismic provisions apply.**

Site Context: Mining & Climate Zone

Three active mineral producers operate within 5 miles: Basalt Rock Co (2 mi, clay), Mt George Pumice Deposit (3.1 mi), and Silva Pumice Quarry (5 mi). While none are oil or gas operations—the nearest plugged well is 7.2 miles distant—the presence of nearby quarries warrants confirmation of blasting protocols, cumulative vibration history, and ground stability in the immediate vicinity. The IECC Zone 3C marine designation imposes moderate heating and cooling loads; building envelope performance will follow prescriptive or performance paths aligned with Title 24 and the California Building Code. **The well-drained soil and zero frost depth support year-round foundation work with standard dewatering protocols.**

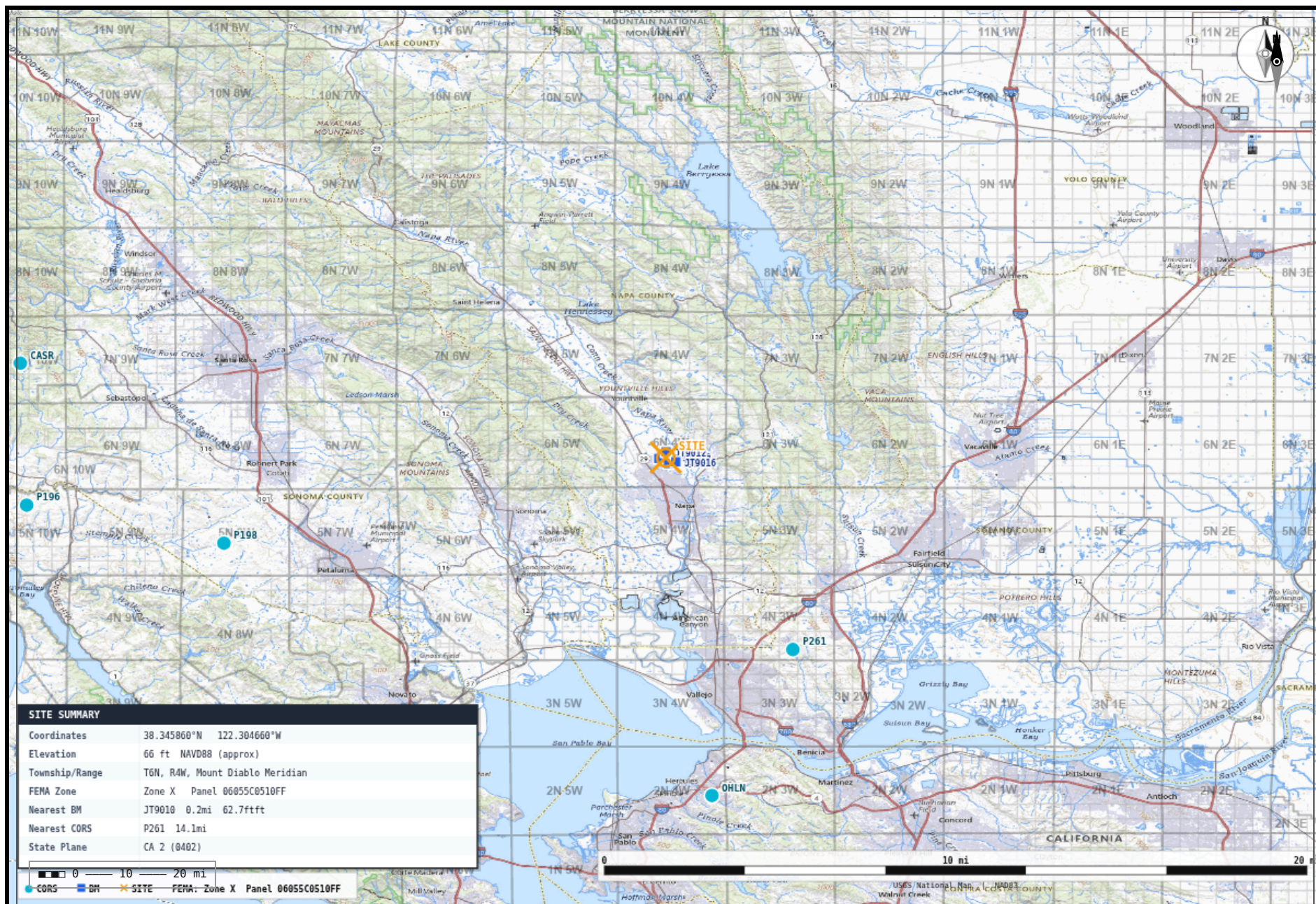
Bottom Line

Initiate subsurface investigation (boring program, SPT sampling) to confirm soil bearing capacity and stratification, and clarify seismic design code applicability with the local jurisdiction and state seismic hazard program.



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Planning aid only — not a substitute for licensed professional review, geotechnical investigation, or code compliance verification.

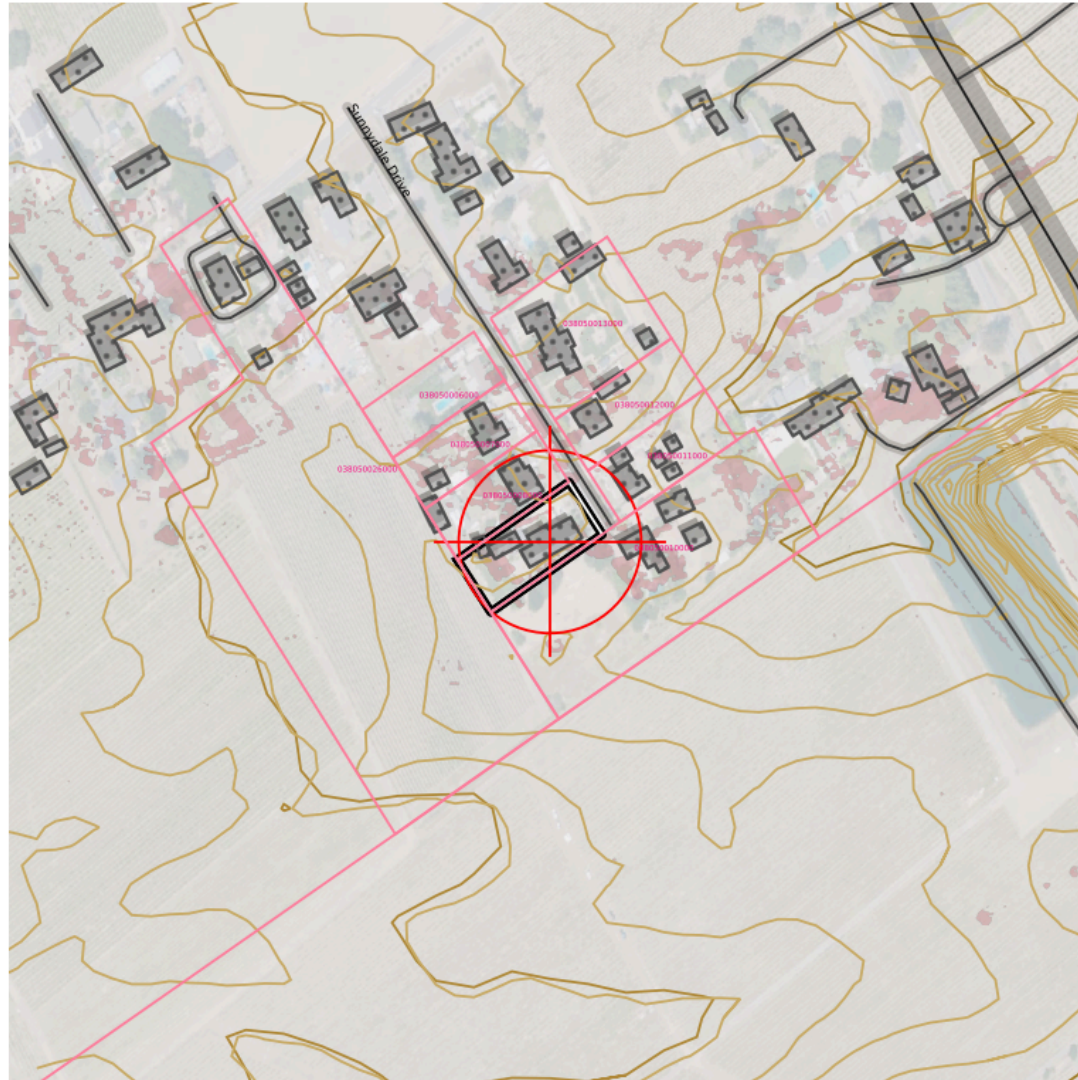


TOPOSCOUT
FIELD MAP

TopoScout Field Map
Lat: 38.34586 Lng: -122.30466
38.345860°N 122.304660°W | Radius: 30 mi

Generated: July 13, 2026
Base: USGS National Map
Generated by TopoScout - outkastdesigns.com

TOPOSCOUT SITE PLAN PREVIEW | \$19 UPGRADE FOR AUTOCAD DXF FILE



INCLUDED LAYERS



0 — 400 ft State Plane · NAD83 · US Feet

- Contours (Index/Major/Minor/1ft)
- PLSS Section/Township/QQ
- Roads Major/Local/Trail + ROW
- Buildings
- Hydro
- Power Lines
- Pipeline
- Railroad
- Benchmarks + Labels
- County Boundary
- OTLS Grid
- Site Info
- Elevation Labels
- 3D Polyline (all geometry)

✂ FIELD RESOURCE LINKS

PRE-MISSION PLANNING

Trimble GNSS Planning Online –

<https://trimblegnssplanningonline.com/>

Satellite visibility, DOP charts, sky plots

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Drone airspace authorization, TFRs, restricted areas

Aloft (Kittyhawk) – <https://www.aloft.ai/>

Field-friendly drone airspace map and LAANC authorization

NOAA Space Weather — Live Kp –

<https://www.swpc.noaa.gov/products/planetary-k-index>

Real-time Kp index, geomagnetic storm alerts

POST-PROCESSING

NGS OPUS – <https://geodesy.noaa.gov/OPUS/index.jsp>

Online Positioning User Service – static GNSS processing

NGS OPUS-RS – <https://geodesy.noaa.gov/OPUSI/index.jsp>

Rapid static processing – sessions as short as 15 minutes

CORS Data Download – <https://geodesy.noaa.gov/CORS/cors-data.shtml>

Raw RINEX observation files from CORS stations

FEMA & FLOOD

FEMA Flood Map Service Center –

<https://msc.fema.gov/portal/home>

FIRM panels, flood zone lookup by address or coordinates

FEMA Elevation Certificate – [https://www.fema.gov/flood-](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

[insurance/work-with-nfip/elevation-certificate](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

Current EC form, instructions, LOMA/LOMR submission guidance

BASE STATION & NETWORK

NOAA CORS Network – <https://geodesy.noaa.gov/CORS/cors-data.shtml>

Continuously operating reference stations, data download

NGS CORS Station Map –

<https://geodesy.noaa.gov/CORS/GoogleMap/CORSmap.shtml>

Interactive map of all CORS stations with status

IGS Station Monitor – <https://www.igs.org/network>

International GNSS Service – global base station health

REFERENCE & CALCULATIONS

NGS Datasheets – <https://www.ngs.noaa.gov/cgi-bin/datasheet.prl>

Benchmarks, control points, passive monuments

NGS Coordinate Conversion (NCAT) –

<https://geodesy.noaa.gov/NCAT/>

NADCON5, datum transforms, state plane conversions

NOAA Magnetic Declination –

<https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml>

Declination calculator for any location and date

NGS Geoid Models – [https://geodesy.noaa.gov/GEOID/](https://geodesy.noaa.gov/GEOID/GEOID18)

GEOID18 and current model downloads, GEOID calculator

DRONE OPERATIONS

FAA Part 107 — Commercial Rules –

https://www.faa.gov/uas/commercial_operators

Full Part 107 regulatory summary for commercial UAS operations

FAA DroneZone – <https://faadronezone.faa.gov/>

Drone registration, operator certificates, waiver management

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Pre-flight airspace check, TFRs, LAANC authorization

FAA Part 107 Waivers –

https://www.faa.gov/uas/commercial_operators/part_107_waivers

Waiver applications for operations beyond standard 107 rules

Aloft — Field Airspace Map – <https://www.aloft.ai/>

Mobile-friendly real-time airspace, LAANC, flight logging

AIR QUALITY & WILDFIRE SMOKE

[AirNow Fire & Smoke Map](https://www.fire.airnow.gov)

<https://www.fire.airnow.gov>

Current fire locations, smoke plumes, PM2.5 readings by location

[AirNow Interactive Map](https://gispub.epa.gov/airnow)

<https://gispub.epa.gov/airnow>

Real-time AQI monitors, hourly NowCast, 48-hr forecast overlays

[NOAA Smoke Forecast](https://airquality.weather.gov)

<https://airquality.weather.gov>

48-hr smoke concentration forecast, HRRR-Smoke model

